- **Problem Description:**

House price prediction has become an important application of machine learning in the real-estate sector. The main objective of this project is to develop a regression-based machine learning model that can estimate the selling price of a house using different property-related features. The dataset used in this project contains 25 attributes such as property area, number of bedrooms, bathrooms, number of floors, year built, property age, distance from the city centre, neighbourhood quality, nearby school rating, hospital distance, parking availability, construction quality, energy efficiency, and other infrastructure facilities.

These factors play a major role in determining the value of a property. By studying past housing data, the machine learning model can understand patterns and relationships between these features and the final house price. In this project, Exploratory Data Analysis (EDA) will first be performed to understand the distribution of the data and the relationship between variables. After that, data pre-processing techniques such as handling missing values, encoding categorical data, and feature scaling will be applied.

Different regression models will then be trained and tested to find the model that provides the most accurate predictions. The final system will help estimate house prices more effectively and can support buyers, sellers, and real-estate agents in making better and more informed decisions.

- **Steps of Implementation:**

1. **Data Loading**
   The dataset is loaded using the Pandas library from a CSV file containing house-related features and prices.
2. **Exploratory Data Analysis (EDA)**
   Basic dataset information such as shape, data types, summary statistics, and missing values are analyzed.
3. **Data Visualization**
   Histograms and correlation heatmaps are used to understand the distribution of house prices and relationships between features.
4. **Data Pre-processing**
   Numerical and categorical columns are separated. Missing values are filled using median or mode values. Categorical variables are encoded using one-hot encoding.
5. **Feature Selection**
   Input features are separated from the target variable (house price).
6. **Train-Test Split**
   The dataset is divided into training and testing sets using an 80:20 ratio.
7. **Feature Scaling**
   StandardScaler is applied to normalize feature values before model training.
8. **Model Training**
   Two regression models are trained:

- Linear Regression
- Random Forest Regressor

9. **Prediction**
   Both models generate predicted house prices on the test dataset.
10. **Model Evaluation**
    The models are evaluated using R^2 score, MAE, and MSE.
11. **Visualization of Results**
    Scatter plots and residual plots are used to compare predicted and actual prices.

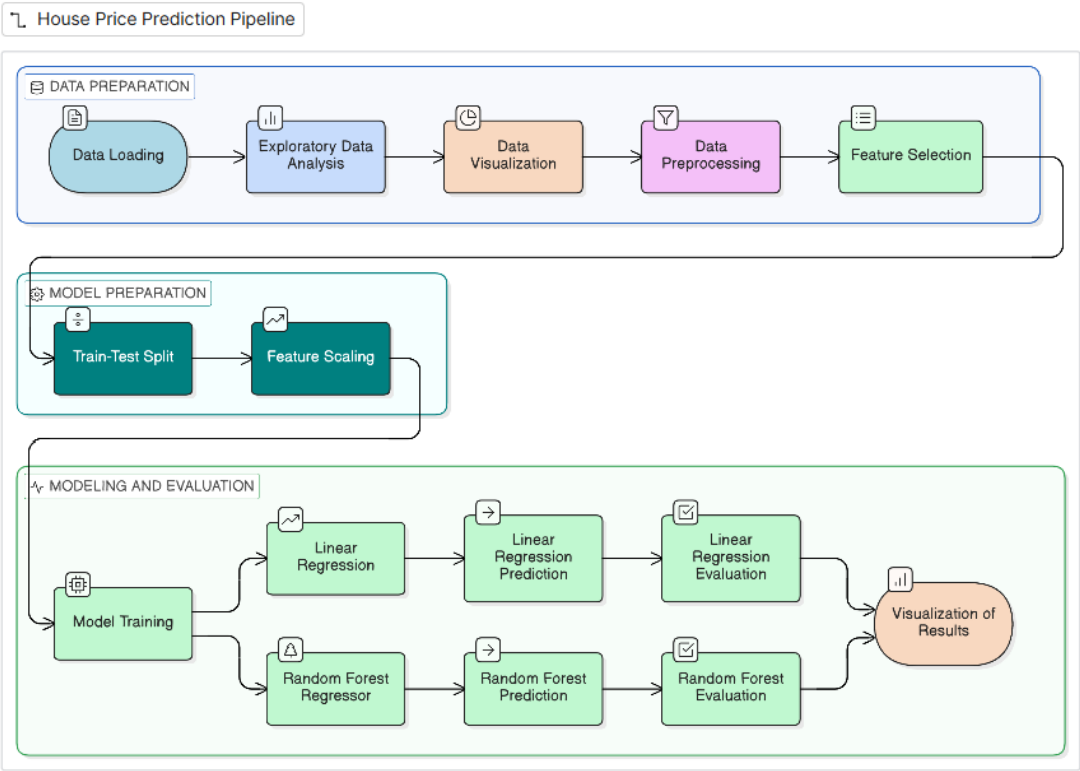

Fig. 1 Sample architecture

- **<u>Expected Modules/ Libraries to be used:</u>**

1. NumPy
Used for numerical computations and handling arrays.

2. Pandas
Used for data manipulation, cleaning, and dataset handling.

3. Matplotlib
Used for data visualization such as graphs and plots.

4. Seaborn
Used for advanced statistical data visualization like heatmaps and pair plots.

5. Scikit-learn
Provides machine learning algorithms such as Linear Regression, Random Forest, and evaluation metrics.

- **<u>Expected Evaluation metrics and Plots to be used:</u>**

R^2 Score
Measures how well the model explains the variance in house prices.

Mean Absolute Error (MAE)
Calculates the average absolute difference between predicted and actual values.

Mean Squared Error (MSE)
Measures the average squared difference between predicted and actual values.

Visualization Plots

- Correlation Heatmap (feature relationships)
- Scatter plot (Actual vs Predicted values)
- Residual plots (model error analysis)
- Distribution plots for feature analysis

These metrics and plots help in evaluating model accuracy and understanding prediction performance.

- **<u>Required topics from the subject:</u>**

  Data Structures (Arrays / Data Handling)
  Used for storing and manipulating dataset features during pre-processing and model training.

  File Handling
  Used to read datasets such as CSV or Excel files for machine learning analysis.

  Functions / Libraries
  Used to perform operations like pre-processing, training models, and evaluating results.

- **Platform Required:**
  Python programming language using **Jupyter Notebook / Google Collab / Anti-Gravity**.

- **Books and Link Sources:**

  Aurelian Geron – *Hands-On Machine Learning with Scikit-Learn & TensorFlow*

  Tom Mitchell – *Machine Learning*

  Scikit-learn Documentation
  https://scikit-learn.org

  Kaggle Datasets
  https://www.kaggle.com